

Assessment of the complete supplied ZIP

Swarm and Evolutionary Computation

Manuscript: Baseline Configuration and Random-Sampling Controls in Metaheuristic Comparisons for Wind Farm Layout Optimization

Assessment date: 4 October 2026.

Material: SWEVO_revision2_full.zip, 223 extracted files; two main PDFs, supplementary PDF, LaTeX sources, generated tables and figures, revision scripts and CSV records.

Verdict: major revision before submission. The research has a credible route to SWEVO as a controlled empirical and methodological study. The supplement materially strengthens the evidence. The submitted archive, however, is not yet a consistent, independently reproducible submission package.

This assessment updates the earlier PDF-only review. It credits experiments and analyses that are now available instead of treating them as missing. It separates confirmed defects from reviewer concerns, and stored-record verification from rerunning the optimization. No manuscript or original ZIP has been edited, and no acceptance probability is assigned.

Dimension	Assessment
Journal relevance	Good potential: swarm/evolutionary comparisons, hybrid components and experimental methodology.
Scientific contribution	Promising if framed around findings that change comparison conclusions; protocol ingredients themselves are established.
Experimental breadth	Substantial: benchmark, controls, budget studies, two site blocks, GA, spacing and exact gradients.
Mathematical correctness	One confirmed exception error in Proposition 1(c)/S1(c); principal PSO stability numbers check out.
Statistical interpretation	Thoughtful sensitivity analyses exist; conditional rankings and extrapolation still require clearer limits.
Reproducibility	Not ready: missing core modules and primary raw runs; rounded coordinates cannot reproduce strict feasibility.
Submission package	Not ready: incompatible versions and unresolved author/deposit declarations.

Read this report as a journal-reviewer-style advisory assessment, not an editorial decision. The highest priorities are archive integrity, reproducibility and correct interpretation, followed by presentation of the strongest comparisons.

What is already addressed by the revision

The 56-page supplement contains substantive answers to several earlier suggestions. Repeating those requests as though no evidence existed would underestimate this revision.

Earlier concern	Evidence now available	Remaining work
GA/evolutionary baseline	S10.2, Tables S56-S58: real-coded GA on all 68 cases, Horns Rev 1 and IEA37; nine-method analysis.	Integrate decisive results in the main comparison and keep statistical families consistent.
Exact-gradient comparison	S10.4, Tables S61-S62: analytic-gradient MS-SLSQP and PSO-SLSQP, 240 runs, gradient checks and charge accounting.	Describe analytic constraint Jacobians and restart strategy; recover missing dependencies.
Laplace step versus evaluation cost	S10.1, Table S55: six new variants isolate re-evaluation, midpoint replacement, scale and related changes.	Keep the limited 12-case power and Holm-adjusted non-significance visible.
Spacing only re-evaluated	S10.3, Tables S59-S60: 6,480 direct optimization runs at 5D and 6D.	Present the reversal at 6D and restricted case set prominently.
Only one measured site	S10.5, Table S63: Lillgrund 16-turbine block at two budgets, 540 runs.	Two blocks remain limited external validation; neither is a full farm design study.
Threshold/dependence robustness	S3: threshold sensitivity, cluster-robust inference, leave-one-cluster-out and related checks.	Use the fixed benchmark as the primary inference target; avoid population-of-farms language.

Scientific strengths worth preserving

The authors explicitly state that PSO-VNS is not a new PSO variant, acknowledge when stronger methods outperform it, distinguish non-significance from equivalence, identify the practical VNS phase as mostly truncated compass search, and discuss model exploitation and feasibility. These features make the paper more credible than a simple ranking exercise.

A promising central message is that configuration, sampling controls, constraint outcomes, objective fidelity and available gradients can change the scientific conclusions of optimizer comparisons. Make those demonstrated changes the contribution. Avoid claiming a general new benchmarking methodology without distinguishing it from established guidance.

SWEVO publishes work on experimental and theoretical aspects of swarm/evolutionary computation and has published methodological/statistical guidance [1-3]. That supports topical fit; it does not settle whether the manuscript clears the journal's novelty and rigor threshold.

A1-A2: resolve versions and missing evidence

A1 - Confirmed: no single consistent submission version

The ZIP contains SWEVO_manuscript.pdf (43 pages) and SWEVO_manuscript_full.pdf (46 pages). The README says the modular and flattened manuscripts produce the same 40-page document; that is inaccurate for this archive. The differences include scientific content, not merely pagination.

For example, optA/sw/08_beyond.tex says an exact-gradient baseline was not tested, and optA/sw/09_robust.tex says the main methods were not re-run at 5D/6D. The fuller manuscript and S10 report those experiments. The modular conclusion also lists some experiments already present in S10 as future work. Rebuilding the modular source would therefore risk losing material.

The fuller PDF and its .tex also disagree in the back matter: the PDF promises an open repository at a TBD Zenodo DOI, while the flattened .tex says data are available on reasonable request. The AI-use declaration differs as well. Select the empirical content of the fuller version as the reconciliation target, verify each difference, then generate every PDF and the flattened source from one authoritative source tree.

A2 - Confirmed: the computational archive is incomplete

Eight supplied CSV families contain 11,640 revision runs. Main Table 2 reports 47,830 optimization runs, excluding 522 diagnostic reruns. Thus the supplied raw CSVs cover the revision experiments but omit 36,190 optimization-run records from the original studies, assuming Table 2's stated accounting. Generated tables and number macros do not replace those records.

The seven revision Python modules import local files not supplied, including mpce_results, mpce_inference_extra, mpce_experiments, iea37_experiments, iea37_model, init_hook, authors_optimizers, authors_objective, wflop_model, feasible_init, extra_baselines, original_vns and hornsrev_model. These are project modules, not packages solved by installing NumPy or SciPy.

Observed failure: running python experiments/rev2_analysis.py --help stops at ModuleNotFoundError: No module named mpce_results. The main analysis and optimizer comparisons cannot be regenerated from this ZIP as supplied.

The README/S9 also reference absent build, reproduction and checking files, including analysis/build_swevo.sh, analysis/build_mpce_paper.sh, analysis/README_reproduce.md and main validation scripts. Recover the original source and run archives, pin dependencies, add a license and a manifest, and verify a small end-to-end reproduction plus regeneration of all results before deposit.

Independent checks of the supplied run data

I read all 140 CSV shards in the eight revision families and verified 11,640 stored records. These checks assess the consistency of supplied records; they do not establish that the optimization code can reproduce them.

CSV family	Runs	Design represented
GA benchmark	2,040	68 cases x 30 seeds
GA Horns Rev 1	60	2 budgets x 30 seeds
GA IEA37	120	2 sizes x 2 budgets x 30 seeds
Exact gradients	240	2 methods x 2 sizes x 2 budgets x 30 seeds
Laplace variants	2,160	6 new variants x 12 cases x 30 seeds
Spacing	6,480	9 methods x 12 cases x 2 spacings x 30 seeds
Lillgrund, 6,030	270	9 methods x 30 seeds
Lillgrund, 30,030	270	9 methods x 30 seeds
Total	11,640	All supplied shard sets complete

Checks that pass

No duplicate run keys were found using method, scenario, size, seed, budget, initialization and spacing where applicable. Calls equal Budget in every supplied record. Ideal - Objective agrees with WakeLoss to a maximum absolute discrepancy of about 6.55×10^{-11} in the recorded objective units. For all 240 gradient records, FunCalls + 3 x GradCalls + Unused equals Budget exactly.

Selected means, best values, feasibility counts and gradient times independently agree with Tables S57-S58 and S61-S63. For example, exact-gradient MS-SLSQP at N=36 and B=6,030 has mean 832,893.28 MWh and best 839,509.33 MWh; PSO-SLSQP at N=36 and B=30,030 has mean 842,556.86 MWh and best 847,780.26 MWh. All 240 gradient records are labelled feasible.

A sensitivity claim that also checks out

The supplement claims every B=6,030 exact-gradient MS-SLSQP run fits within 30,030 charged evaluations for c_g up to 19. Checking individual records rather than just the averages supports this: at c_g=19, costs range from 29,038 to 29,486 for 16 turbines and 29,245 to 29,550 for 36 turbines. This arithmetic claim is valid for the supplied traces; it is not an independently rerun charge-sensitivity experiment.

The original eight-method benchmark, coefficient sweep, original control contrasts, finer-direction re-evaluations and their reported automated checks remain unverified computationally because the necessary raw files and modules are absent.

A3: coordinate precision breaks exact replay

Confirmed archival defect. The CSV serialization rounds benchmark/IEA coordinates to 0.001 m and Lillgrund coordinates to 0.01 m. The stated feasibility tolerance is 0.000001 m. Rounded representations do not preserve that tolerance for active constraints.

I reconstructed stored coordinates, computed pairwise minimum spacing and circular or supplied Lillgrund boundaries, and applied the stated 10^{-6} m tolerance. The following rows are labelled feasible in the archive but do not pass the same geometric criterion when reconstructed from the stored coordinates.

Family	Labelled feasible	Rounded layouts failing spacing or boundary
Exact gradients	240	236
Laplace variants	1,922	78
Spacing 5D/6D	4,259	229
Lillgrund 6,030	77	17
Lillgrund 30,030	232	73

For the exact-gradient family, 236 violate a circular boundary and 32 violate minimum spacing; these sets overlap. The largest rounded spacing shortfall is about 0.000936 m, and the largest boundary excess about 0.000628 m. On Lillgrund, the largest rounded spacing shortfall is about 0.00781 m. These are small distances but exceed the explicitly claimed tolerance.

Objective replay can change materially on the top-hat model

The self-contained `rev2_site_model.py` permits an independent AEP evaluation without the absent optimizer modules. Most rounded Lillgrund layouts reproduce AEP closely. However, 15 feasible layouts in the 30,030-budget family differ by more than 0.001 GWh. The largest discrepancy is PSO seed 18: stored AEP 113.473839 GWh, reconstructed-coordinate AEP 113.248767 GWh, a difference of 0.225072 GWh. PSO-VNS seed 15 differs by 0.146511 GWh.

The model has discontinuous top-hat membership tests, so rounding can switch which turbine pairs are in a wake. This is a plausible explanation supported by the model structure, not a proof of the original full-precision trajectories. The mismatch is directly observable; it should not be dismissed as ordinary last-digit printing noise.

Required correction and verification

Recover full-precision coordinates from original binary records or rerun deterministically after recovering code. Store float64 arrays or round-trip decimal representations (e.g., 17 significant digits), with coordinate units and model/version metadata. Keep rounded coordinates only for printed tables. Recompute and check feasibility, AEP and summary tables from the archival representations.

Interpretation limit: these checks do not prove that the original optimization outputs were infeasible, nor that the comparative conclusions reverse. They prove that the current archived coordinates cannot validate those original claims at the stated precision.

A4: mathematical correctness and scope

Confirmed exception error in Proposition 1(c) and S1(c)

The proposition allows c_1 and c_2 to be nonnegative, with c_1+c_2 positive, but asserts that the limiting variance is zero if and only if $p=g$. Its own expression $q = c_1^2 c_2^2 (p-g)^2 / [6(c_1+c_2)^2]$ has additional zero cases when either acceleration coefficient is zero.

Correct statement under the existing assumptions: the limit is zero if and only if $c_1 c_2 (p-g)=0$. Alternatively, assume $c_1>0$ and $c_2>0$ throughout the proposition, and retain the existing $p=g$ equivalence.

A counterexample is $w=0$, $c_1=0$, $c_2=1$ and p different from g . The order-2 condition is satisfied: $\text{varrho}(1)=2/3$ and the second-moment spectral radius is $1/3$. The particle contracts towards g and has zero limiting variance although p differs from g . Correct both the main and supplementary statement, their explanations, and any copied version.

The principal stability classification remains supported

Independently forming the stated second-moment matrix gives spectral radius 1.123652 for $w=0.7$, $c_1=c_2=2$ and 0.944179 for $w=0.7298$, $c_1=c_2=1.49618$. These agree with the rounded reported values 1.12 and 0.94. The exception error does not overturn the empirical comparison of those two positive-coefficient settings.

The distinction between order-1 and order-2 stability is carefully explained, and the supplementary proof gives the forcing and covariance terms rather than silently discarding them. The work correctly states that the general condition is a restatement of existing analysis, rather than claiming an entirely new stability theorem.

Maintain the stated limitations of the theorem

The analysis assumes fixed personal/global bests and inactive bounds. Position clipping with retained velocity invalidates the recurrence used in that calculation. The paper acknowledges this and tests coefficient changes and clamping. Present the theorem as an explanation of the unconstrained stagnation dynamics and the sweep as empirical evidence. It does not prove that variance instability causes every observed failure in the bounded WFLOP implementation.

Likewise, the VNS statements concern the implemented truncated local search at the specified budget. The controls and diagnostics support conclusions about that implementation, rather than all possible VNS or salp-swarm variants. Preserve the restricted language in the abstract, conclusions and highlights.

A5-A6: reliability and statistical interpretation

A5 - Qualification plus conditional quality is not all-run reliability

The ranking qualifies a method with at least 15 feasible runs out of 30, then orders qualified methods by the mean quality of their successful runs. This is an explicit, defensible descriptive rule, but it can put a less reliable method above a method feasible on every run. The threshold sensitivity already reported is useful and should be retained.

Lillgrund at B=6,030	PSO-VNS	MS-SLSQP
Feasible runs	17/30	30/30
Conditional mean AEP	113.0740 GWh	112.8653 GWh
Qualified-mean rank	1	2
Reported paired feasibility-first test	Inferior to MS-SLSQP	Better, Holm p=0.010

This apparent contradiction is explained in S10.5, so the authors have not hidden it. Make it equally visible in the main display. Report success probability with uncertainty, conditional feasible quality, and an all-run paired outcome side by side. Avoid referring to qualified-mean rank alone as the best operational optimizer.

The composite run-level endpoint orders infeasible runs by minimum spacing while ignoring boundary violation. A point far outside a farm can have large spacing. Use a documented normalized measure of both violations, or show a sensitivity analysis, and make clear that the case-mean signed-rank test uses constructed maximal differences when only one method qualifies.

A6 - Treat the fixed benchmark as the primary target

The manuscript already separates seed-, case- and cluster-level inference, reports failure to establish cluster-level equivalence for PSO-VNS versus PSO, and provides multiple dependence checks. Those are strengths. The six dataset-radius groups remain designed synthetic scenario groups, not a random sample of six real farms; a robust standard error cannot create representativeness.

Explicitly state the estimand and weights. The groups have 9, 11 or 14 cases, so equal case weighting differs from equal group weighting. Describe joint resampling of paired seeds and preservation of any cross-case seed dependence. Label case resampling as exchangeability-based sensitivity and cluster inference as sensitivity to grouped dependence, unless a target population and sampling argument are supplied.

S6.4 explicitly says its reported bootstrap p value is a percentile-interval inversion, not a null-boundary-calibrated p value. Keep that qualification with the main results. The TOST intersection-union size argument is valid for calibrated component tests; it does not itself calibrate this bootstrap-tail procedure or eliminate multiplicity across different algorithm pairs. Use confidence-interval-based equivalence language or provide the calibration and pair-family policy.

A7-A8: integrate and describe stronger baselines

A7 - Decisive competitors belong in principal displays

A GA analysis already exists in Table S56 with the nine-method pool; the issue is visibility and consistency, not its absence. The GA ranks third on the benchmark and has the strongest Horns Rev 1 mean at 30,030 evaluations among the methods compared there. With 36 IEA37 turbines it is strongest among the tested metaheuristics, while the exact-gradient methods are stronger still.

Retain the original eight-method analysis as the historical primary analysis if useful, but show the augmented comparison prominently. Do not compare average ranks calculated with different method pools as though they have the same scale. Adding rows requires the corresponding ranks, tests and multiplicity families, not manual reuse of the old values.

N / budget	Exact MS-SLSQP mean	PSO-SLSQP mean	Mean runtime, seconds
16 / 6,030	407,720.99	407,891.39	4.27 / 5.77
16 / 30,030	409,976.31	413,179.21	34.70 / 23.71
36 / 6,030	832,893.28	832,404.02	15.95 / 12.41
36 / 30,030	838,852.89	842,556.86	85.35 / 56.62

AEP is in MWh; these numbers and times were independently recomputed from the supplied records. The main/S10 tables show these methods outperforming the metaheuristics. This is central to the paper's methodological message and should be discoverable directly from its primary IEA37 table.

A8 - The gradient experiment changes more than one derivative

rev2_gradient.py explicitly supplies analytic constraint Jacobians as well as analytic objective gradients; the original method forward-differences both. The supplement mentions this, but isolated statements attributing the improvement only to the objective gradient are too narrow. Describe the comparison as exact-gradient SLSQP with analytic constraint Jacobians versus the finite-difference implementation.

The PSO-SLSQP hybrid also includes an alternating restart policy using perturbed incumbents and swarm personal bests. Sharing Phase 1 supports comparison of the whole second-stage procedures, not a conclusion that the derivative alone explains the improvement. An objective-gradient-only control is needed only if that causal attribution is retained.

Charge accounting is internally correct and conservative relative to the measured ratios. Constraint/Jacobian evaluation and solver work are uncharged, however. The runtime discussion already recognizes substantial SLSQP overhead and different timing batches. Present charged-budget and wall-clock conclusions separately; a same-machine timing experiment would strengthen any claim of general speed superiority.

A9-A10: model transfer and constraint fairness

A9 - Re-evaluation and optimization answer different questions

The finer-direction, Gaussian and power-curve experiments mostly re-evaluate final layouts generated under the original model. They test whether the archived solutions and their ordering transfer to another evaluator. They do not establish which optimizer would win if each searched that alternative objective. The full revision appropriately adds direct 5D/6D optimization; credit that distinction.

The spacing study is informative because the SSA-versus-disc-control conclusion holds at 5D but changes at 6D. The geometrically impossible radius-500/N=10 cases are excluded from comparable feasible-case summaries, leaving 10 cases and, for some contrasts, 8. Present that restriction and reversal where the principal conclusion is made, not only as a supplementary qualification.

A small direct optimization study at 1-degree resolution would strengthen claims about finer roses. It is recommended if the authors want such claims, but it is not an automatic requirement if they consistently state that those results concern re-evaluation of archived solutions. The IEA37 Gaussian study already provides a separate optimization comparison on a different model.

Site studies are optimizer tests, not validated engineering designs

Horns Rev 1 and Lillgrund use 16-turbine blocks and omit the rest of each farm. The Lillgrund model uses seven months of wind measurements, a 3D minimum spacing and 5-degree directions. The benchmark uses idealized circular sites, a top-hat formulation and a simplified power curve. The paper candidly discusses upstream-deficit and local-thrust-coefficient conventions and model exploitation; retain those explanations.

I independently reproduced the supplied Lillgrund model's installed-block AEP of 116.096345 GWh and ideal AEP of 139.401961 GWh. The optimization records show neither budget improves on the installed block on average. This is useful negative evidence, but does not constitute validation against measured farm production, loads or economic performance.

A10 - Constraint handling is part of the measured algorithm

A common very large penalty mixes boundary violations in square metres with spacing violations in metres, and box clipping does not repair the circular farm constraint. SLSQP instead uses explicit constraints. This protocol can be reproducible while still favoring different move structures. Poor DE feasibility therefore need not describe the capability of DE as an algorithm family.

The clamping, feasible-start and sampling controls already probe some of these effects. A compact sensitivity study with normalized violations and a shared feasibility rule would materially strengthen broader claims. If no such study is added, restrict conclusions to the exact configurations and handling used; avoid implying that one default represents the whole evolutionary family. CMA-ES or adaptive DE is a useful extension, not a universal mandatory extra experiment for this focused study.

A11-A12: declarations and presentation

A11 - Resolve every visible placeholder and factual declaration

The 46-page main PDF still prints red TBD items on pages 41-42 for CRediT roles, competing-interest confirmation, the repository DOI, AI-use confirmation, funding and acknowledgment. The cover letter also has a date placeholder. These are direct submission blockers; deleting confirmation markers is appropriate only after the corresponding factual declarations have been verified by the authors.

Repository statements must match actual access. The PDF promises all methods, evaluators, per-run records and analyses, while the flattened source says reasonable request and the ZIP lacks much of the promised material. Choose a truthful policy, complete the deposit and use its real identifier consistently across main PDF, source, declarations, cover letter and reproduction instructions. This recommendation follows the paper's central reproducibility claim; it is not asserted here as a verified journal repository mandate.

Confirm the AI disclosure reflects the actual workflow, including drafting/editing if that occurred. The current source and compiled PDF describe different scopes. Author roles, funding, interests and AI-use facts cannot be inferred from code or supplied prose and should not be invented to make the package look complete.

A12 - Reduce review burden without discarding decisive evidence

The combined main/supplement exceeds 100 pages in the fuller version. Length alone is not a confirmed journal-rule violation, but an editor needs a clear argument. Several wide tables use very small type, notably multi-level equivalence and cross-method summaries. Move secondary details to the supplement and give central results readable displays with the comparator, effect, uncertainty, feasibility and scope.

Rebuild the abstract and cover letter around the final contribution and the strongest findings, including where they fail: baseline configuration reverses conclusions; sampling controls expose limited component value at the tested settings; small average gains are heterogeneous; gradient baselines dominate on IEA37. Avoid an abstract primarily cataloguing every test or an implication that PSO-VNS is universally best.

Use a short research-question-to-evidence map. Separate primary analyses from post hoc additions, preserving the already disclosed exploratory margin, subgroups and density pattern. The fuller text is generally candid; the priority is consistency and placement, followed by line editing, not a wholesale stylistic rewrite.

Build verification boundary

Local compilation attempts stopped because the available TeX installation lacks elsarticle.cls for the main and algorithm.sty for the supplement. These are standard dependencies, so the failures do not establish defective LaTeX source. After reconciling versions, compile in a complete TeX environment, check cross-references and citation numbering, and visually inspect every final page. The stale README and absent build script still require correction independently of the local dependency limit.

The live ScienceDirect author-guide page was inaccessible during this review. The package's own policy checklist should therefore be treated as an author claim until checked against the submission system. I do not certify its page/abstract limits, anonymity settings or required upload item types.

Prioritized revision and implementation tracker

P0: fix before submission. P1: central scientific interpretation or evidence presentation. P2: strengthening work, dependent on the breadth of claims. "With supplied ZIP" means an honest implementation can be made using present material; it does not mean the revision has been performed.

ID / priority	Action	Can be completed from this ZIP?
A1 / P0	Reconcile main versions, declarations and source; rebuild one consistent release.	Content reconciliation yes; successful final build needs complete TeX dependencies.
A2 / P0	Recover all local modules, original 36,190 run records and provenance/checking scripts.	No. The missing scientific material must be recovered or rerun.
A3 / P0	Archive full-precision layouts and validate replay at the stated tolerance.	No. Discarded decimal precision cannot be recovered from current CSVs.
A4 / P0	Correct Proposition 1(c)/S1(c) and dependent wording.	Yes. Exact correction is established analytically.
A5 / P1	Display reliability, conditional quality and all-run outcomes together.	For revision datasets yes; original complete analyses need missing raw data.
A6 / P1	Clarify inference target, weights, bootstrap equivalence and multiplicity.	Wording yes; recalibration/recomputation needs original pipeline and data.
A7 / P1	Promote GA and gradient comparisons into primary tables and conclusions.	Existing results can be integrated; original augmented statistics cannot be independently regenerated yet.
A8 / P1	Describe analytic constraint Jacobians, hybrid restarts and cost limits.	Yes for accurate description; isolating causal controls requires runs/code recovery.
A9 / P1-P2	Separate transfer re-evaluation from optimization; expose 6D reversal.	Yes for scope correction; finer-rose optimization requires new runs.
A10 / P2	Test normalized penalty/feasibility handling or restrict broad claims.	Scope wording yes; a meaningful sensitivity experiment needs complete code.
A11 / P0	Finalize author declarations and a truthful accessible deposit.	Requires author facts and recovered deposit contents; cannot be inferred.
A12 / P1	Update README, cover letter and main argument; improve dense displays.	Yes for editorial revisions; release validation depends on A1-A3.

The earlier estimate that several suggestions could be completed assumed a complete source/data bundle. This ZIP contains substantial revision evidence but not that complete bundle. It would be misleading to promise a fully verified revised ZIP without first recovering missing modules, original runs and full-precision coordinates.

Readiness gates and reviewer recommendation

Minimum evidence for changing the verdict to ready to submit

- 1. One consistent release.** Main manuscript, supplement, modular/flattened sources, declarations and cover letter agree. The fuller revision findings survive rebuilding; the README documents the actual build and file layout.
- 2. Recoverable computation.** All project modules and original/revision raw data are present with environment information. A fresh setup runs a smoke reproduction, regenerates tables and figures from records, and checks selected optimization/evaluator outputs. The repository statement names an accessible, complete deposit.
- 3. Faithful archival layouts.** Full-precision coordinates reproduce feasibility and objectives within declared numerical tolerances. Rounded presentation files are clearly separated from the authoritative records. Any summary affected by corrected records is regenerated.
- 4. Correct and bounded claims.** The variance exception is fixed. Conditional feasibility rankings, the fixed benchmark target and bootstrap procedure are accurately named. Claims about gradients cover both objective and constraint derivatives. Stronger baselines and the 6D reversal are visible in principal results.
- 5. Finished submission material.** Author-confirmed declarations contain no placeholders. Final PDFs have readable tables, resolved references, consistent captions and an abstract/cover letter that match the final study.

Reviewer recommendation: major revision before submission. The study is a plausible SWEVO candidate and the supplement substantially improves its scientific case. Its present archive and version state would undermine the central claim of a fair, reproducible comparison. The essential next step is to repair and validate that evidence chain, not merely polish prose.

Limits of this audit

Inspected: archive inventory, both main PDFs and sources, 56-page supplement, revision scripts, all supplied CSVs and selected rendered scientific/declaration pages. Independently checked: record identities and budgets, selected numerical summaries, coordinate geometry, the supplied Lillgrund evaluator, gradient charge sensitivity arithmetic and PSO spectral values. Not independently reproduced: original optimization runs, original inference/re-evaluation pipeline, claimed 200-plus checks, PyWake/JAX comparisons or complete LaTeX builds. The original ZIP was preserved unchanged.

Primary journal context and methodological references

- [1] Elsevier, Swarm and Evolutionary Computation description within Decision sciences journals. Retrieved 4 October 2026. [Publisher description](#). Used for topical scope only; the live author guide was not retrievable.
- [2] A. LaTorre et al., A prescription of methodological guidelines for comparing bio-inspired optimization algorithms. Swarm and Evolutionary Computation 67 (2021), 100973. [doi:10.1016/j.swevo.2021.100973](#). Supports the assessment of benchmark selection, comparator choice and principled experimental validation.
- [3] J. Carrasco et al., Recent trends in the use of statistical tests for comparing swarm and evolutionary computing algorithms: practical guidelines and a critical review. Swarm and Evolutionary Computation 54 (2020), 100665. [Author preprint](#). Used as methodological context for test assumptions and interpretation.